

Entity Name	Table Name	Screen/Module	Attribute Name	Data Type	Source of Value (UI / MLDataProcessor /	Nullable	Default Value	Description / Why Needed
User	users	Login	id	Long	System (Auto-generated)	No	Auto	Primary key
		Admin	keycloakid	String	System (Auto-generated-through-	No	Auto	Login
			fullName	String	UI Input	No		User Details
			email	String	UI Input	No		Login, Marketing
			mobileNumber	String	UI Input	No		Contact, Marketing
			username	String	System (Auto-generated-through-keycloak)	No	email	Login
			password	String	UI Input	No		Login
			userStatus	enum	System (Auto-generated-through-	No	Enum{ACTIVE,PENDING,INACTIVE}	User Details
			designation	String	UI Input	No		User Details
			address	String	UI Input	Yes		User Details
			pincode	String	UI Input	Yes		User Details
			tenantId	Long	System (tenant context)	No		Associates user with a tenant
Client	client	Client Master	id	Long	System (Auto-generated)	No	Auto	Primary key
		Activity	clientName	String	UI Input	No	None	Name of the client
		Dashboard	userStatus	enum	System / UI	No	ACTIVE	Status of client user
			contactName	String	UI Input	Yes		Contact person name
			user	Long	Dropdown / FK	Yes		Linked user (owner)
			status	enum	System / UI	Yes		Active/inactive status
			registeredUnder	String	UI Input	Yes		Registration info
			address1	String	UI Input	Yes		Street address line 1
			address2	String	UI Input	Yes		Street address line 2
			address3	String	UI Input	Yes		Street address line 3
			city	String	UI Input	Yes		City
			state	String	UI Input	Yes		State
			country	String	UI Input	Yes		Country
			panCard	String	UI Input	Yes		Tax ID / PAN
			ownerName	String	UI Input	Yes		Name of client owner
			phoneNumber	String	UI Input	Yes		Primary phone number
			secondaryPhoneNumber	String	UI Input	Yes		Secondary phone number
			singlePointOfContact	String	UI Input	Yes		SPOC for client
			email	String	UI Input	Yes		Primary email
			secondaryEmail	String	UI Input	Yes		Secondary email
			bankName	String	UI Input	Yes		Bank for payments
			accountNumber	String	UI Input	Yes		Bank account number
			accountType	String	UI Input	Yes		Account type
			ifscCode	String	UI Input	Yes		Bank IFSC code
			gstIn	String	UI Input	Yes		GST identification
			branch	String	UI Input	Yes		Branch location
			zipCode	String	UI Input	Yes		Postal code
			clientCode	String	UI Input / System	No	Auto	Unique code for client
			tenantId	Long	System (tenant context)	No		Associates user with a tenant
Tenant	tenant	Tenant (System)	id	Integer	System (Auto-generated)	No	Auto	Primary key
			name	String	UI Input	No		Tenant name
			description	String	UI Input	Yes		Tenant description
			legalName	String	UI Input	No		Legal name of tenant
			address	String	UI Input	Yes		Tenant address
			contactNumber	String	UI Input	Yes		Contact phone number
			contactEmail	String	UI Input	Yes		Contact email address
			logoUri	String	UI Input	Yes		URL to tenant logo
			primaryColor	String	UI Input	Yes		Primary brand color
			secondaryColor	String	UI Input	Yes		Secondary brand color
Project	project	Project	id	Long	System (Auto-generated)	No	Auto	Primary key
		Activity	projectName	String	UI Input	Yes		Name of the project
			industryType	enum	UI Input	Yes	Enum {Retail, logistics, other}	Type of client project (ClientProjectType enum)
			primaryContactName	String	UI Input	Yes		Primary contact person name
			primaryContactNumber	String	UI Input	Yes		Primary contact phone number
			primaryEmailId	String	UI Input	Yes		Primary contact email
			status	enum	UI Input	Yes		Project status (Status enum)
			client	Long	Dropdown / FK	Yes		Foreign key to Client entity
			dataSchemaRequirement	String	MLDataProcessor (Validation process)	Yes		Schema requirements for the data
			tenantId	Long	System (tenant context)	No		Associates user with a tenant
Activity	activity	Dashboard	id	Long	System (Auto-generated)	No	Auto	Primary key
		Activity	clientId	Long	Dropdown / FK	No		Foreign key to Client entity
			projectId	Long	Dropdown / FK	No		Foreign key to Project entity
			name	String	UI Input	No		Activity name
			description	String	UI Input	Yes		Activity description
			priority	enum	UI Input	No	Enum{HIGH,MEDIUM,LOW}	Activity priority (Priority enum)
			forecastCount	Integer	System / UI	Yes	0	Number of forecasts associated

			tenantId	Long	System (tenant context)	No		Associates user with a tenant
UploadedData	uploaded_data	Data Upload	id	Long	System (Auto-generated)	No	Auto	Primary key
			fileName	String	System / UI	No		Name of uploaded file
			url	String	System (File storage)	No		URL/Path to uploaded file
			timeStamp	DateTime	System (Auto-generated)	No	Current timestamp	Upload timestamp
			uploadDataType	enum	UI Input	No	Enum{RAW, PREPROCESSED}	Type of uploaded data (UploadDataType enum)
			dataSchemaRequirement	String	MLDataProcessor (Validation process)	Yes		Validated Schema requirements for the data
			validationResults	String	MLDataProcessor (Validation process)	Yes		Results of data validation
TransformedData	transformed_data	Data Upload	id	Long	System (Auto-generated)	No	Auto	Primary key
			dataOverview	JSON	MLDataProcessor (Processing)	Yes		JSON containing data overview information
			dataStatistics	JSON	MLDataProcessor (Processing)	Yes		JSON containing statistical analysis of data
			dataSample	JSON	MLDataProcessor (Processing)	Yes		JSON containing sample data records
ExploratoryDataAnalytics	exploratory_data_analytics	Data Analysis	id	Long	System (Auto-generated)	No	Auto	Primary key
			dataOverview	JSON	MLDataProcessor (Processing)	Yes		JSON containing data overview information
			dataQuality	JSON	MLDataProcessor (Processing)	Yes		JSON containing data quality metrics
			temporalAnalysis	JSON	MLDataProcessor (Processing)	Yes		JSON containing temporal analysis results
			distributionalAnalysis	JSON	MLDataProcessor (Processing)	Yes		JSON containing distributional analysis
			categoricalAnalysis	JSON	MLDataProcessor (Processing)	Yes		JSON containing categorical data analysis
			relationshipCorrelations	JSON	MLDataProcessor (Processing)	Yes		JSON containing relationship and correlation analysis
Models	models	Model Selection	id	Long	System (Auto-generated)	No	Auto	Primary key
			modelName	String	System / Admin UI / SQL	No		Name of the model
			modelDescription	String	System / Admin UI / SQL	Yes		Description of the model
			modelType	enum	System / Admin UI / SQL	No	Enum{STATISTICAL, MACHINE_LE	Type of model (ModelType enum)
			modelIcon	String	System / Admin UI / SQL	Yes		Icon URL or identifier for the model
			complexity	String	System / Admin UI / SQL	Yes		Model complexity level
			accuracy	String	System / Admin UI / SQL	Yes		Model accuracy metrics
			estTime	String	System / Admin UI / SQL	Yes		Estimated execution time
			bestFor	String	System / Admin UI / SQL	Yes		What the model is best suited for
			isRecommended	Boolean	System / Admin UI / SQL	Yes	false	Whether model is recommended
Features	features	Feature Selection	id	Long	System (Auto-generated)	No	Auto	Primary key
			features	JSON	UI Input	Yes		JSON containing feature data and metadata
			modelID	Long	UI Input	No		Foreign key to Models entity
Aggregation	aggregation	Aggregation (Forecast)	id	Long	System (Auto-generated)	No	Auto	Primary key
			geographicAggregationLevels	JSON	UI Input	Yes		Geographic aggregation levels: Individual, Regional,
			productAggregationLevels	JSON	UI Input	Yes		Product aggregation levels: Individual, Product
			temporalAggregationLevel	JSON	UI Input	Yes		Temporal aggregation levels: Daily, Weekly, Monthly
			minimumThresholdDropDown	JSON	UI Input	Yes		Minimum threshold dropdown options
			geographicAggregationDetails	JSON	System / MLDataProcessor / UI	Yes		Detailed geographic aggregation configuration
			productAggregationDetails	JSON	System / MLDataProcessor / UI	Yes		Detailed product aggregation configuration
			temporalAggregationDetails	JSON	System / MLDataProcessor / UI	Yes		Detailed temporal aggregation configuration
			forecastHorizon	JSON	System / MLDataProcessor / UI	Yes		Forecast horizon configuration
DateRangeConfiguration	date_range_configuration	Date	id	Long	System (Auto-generated)	No	Auto	Primary key
			trainingDateRange	JSON	UI Input	Yes	DateRangeObject	Date range configuration for training data
			testingDateRange	JSON	UI Input	Yes	DateRangeObject	Date range configuration for testing data
ForecastResult	forecast_result	Forecast Results	id	Long	System (Auto-generated)	No	Auto	Primary key
			modelResultAnalysis	JSON	MLDataProcessor (Processing)	Yes		JSON containing model result analysis
			modelPerformanceSummary	JSON	MLDataProcessor (Processing)	Yes		JSON containing model performance summary
			historicalModelFit	JSON	MLDataProcessor (Processing)	Yes		JSON containing historical model fit data
ValidationResults	validation_results	Data Upload	id	Long	System (Auto-generated)	No	Auto	Primary key
			totalRows	Integer	MLDataProcessor (Processing)	Yes		Total number of rows in dataset
			missingValues	Integer	MLDataProcessor (Processing)	Yes		Count of missing values detected
			detectedOutliers	Integer	MLDataProcessor (Processing)	Yes		Count of outliers detected
			outlierColumns	String	MLDataProcessor (Processing)	Yes		Columns containing outliers
			additionalColumns	String	MLDataProcessor (Processing)	Yes		Additional columns information
Forecast	forecast	Dashboard	id	Long	System (Auto-generated)	No	Auto	Primary key
		Forecast Creation	uploadedDataId	Long	FK	Yes		Foreign key to UploadedData entity
		Forecast	activityID	Long	FK	No		Foreign key to Activity entity
			status	enum	MLDataProcessor (Processing)	No	Enum{DRAFT, RUNNING, ACTIVE, C	Forecast status (ForecastStatus enum)
			transformedDataId	Long	FK	Yes		Foreign key to TransformedData entity
			forecastType	enum	MLDataProcessor (Processing)	Yes	Enum{DEMAND, FORECAST, INVE	Type of forecast (ForecastType enum)
			modelId	Long	FK	Yes		Foreign key to Models entity
			featureId	Long	FK	Yes		Foreign key to Features entity
			aggregationId	Long	FK	Yes		Foreign key to Aggregation entity
			dateRangeConfigId	Long	FK	Yes		Foreign key to DateRangeConfiguration entity
			resultId	Long	FK	Yes		Foreign key to ForecastResult entity
			exploratoryDataAnalyticsId	Long	FK	Yes		Foreign key to ExploratoryDataAnalytics entity
			tenantId	Long	System (tenant context)	No		Associates user with a tenant